

INFO1113 Gremlins Game Report

Explanation of object-oriented design decisions

On setup the program creates the level classes which will allow it to access the levels at will and will save the state of the level even after the level has been completed.

Updating the screen is mainly handled in App. But each level is handled through the level for example at the start of the level the program calls the level and from then on, the app will only change the state of the current level.

Each Level object holds their respective tile setup, entity setup, powerup setup, projectile setup.

Class hierarchy

Tile is a parent class to Brickwall, Stonewall and Exit as it joins them together by both being a tile. And a non-moving object.

Entity is a parent class to Gremlin and Wizard as they move around the course.

Projectile is a parent class to Fireball and Slime as they are both shot from the wizard and gremlins respectively. Making a projectile parent class allows easier implementation of more projectiles.

Powerup is a parent class to Invincibility as is it will give the wizard an advantage. Making a powerup parent class allows easier implementation of more powerups.

Abstract classes

The abstract classes are Tile, Entity, Projectile and Powerup. I have used abstract classes to group objects together and to make the program simpler to understand.

Tile

The parent tile is used to hold the x and y values of the tile and handles breaking (in case a new tile is added that also breaks/explode). Breaking could have been handled by only Brickwall as neither Stonewall nor Exit break.

Entity

The Entity class is used to hold the location of the entity at any point in time and handles the entity's cooldowns. This allows a simpler implementation of Wizard and Gremlin.

Projectile

The Projectile class is used to handle the location and direction of the projectile.

Powerup

The powerup class was used to group additional powerups that can be added in future. And handles the location and powerup cooldown of the powerup.